

Theater of Future Puppetry: An Immersive Puppeteering Experience Based on Hands Tracking and Gestures Recognition

Wei-Chen Yen

wilsonnosliw885@gmail.com
Institute of Applied Arts
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan (R.O.C.)

Jen-Kai Liu

kenny0408237@gmail.com
Institute of Applied Arts
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan (R.O.C.)

Jia-Jiu Peng

George.peng.tw@gmail.com
Institute of Applied Arts
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan (R.O.C.)

Hsiao-Yu Luo

hsiaoyu222@gmail.com
Institute of Applied Arts
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan (R.O.C.)

Ping-Hsuan Han

pinghsuan.han@gmail.com
Department of Interaction Design
National Taipei University of
Technology
Taipei, Taiwan (R.O.C.)

Chun-Cheng Hsu*

cchsu@nycu.edu.tw
Institute of Applied Arts
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan (R.O.C.)

ABSTRACT

There are many studies on hand tracking and interaction with virtual puppets in the immersive experience. This research presents an advanced gesture recognition system called Theater of Future Puppetry. Unlike systems that only use hand tracking and hand skeleton manipulation of puppets, this system can recognize specific gestures used in traditional Budaixi (Taiwanese glove puppetry) and interact with them. The gestures were designed based on discussions with traditional Budaixi masters. Through this system, users can learn and understand the knowledge and cultural background of Budaixi puppet manipulation.

CCS CONCEPTS

• **Human-centered computing** → *Virtual reality*.

KEYWORDS

Hand Tracking, Gesture Recognition, Puppetry, Budaixi, Virtual Reality, Interaction

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1 INTRODUCTION

Budaixi is a traditional puppetry form in Taiwan. We are devoted to digitizing the puppeteering experiences in virtual reality (VR).

*Corresponding Author

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Manipulation of virtual puppets can be facilitated in many ways, Lin et al. [Lin et al. 2018] use Leap Motion as a hand-tracking system to manipulate the puppets and detail how to connect the hand skeleton to a virtual puppet avatar and successfully control it. Nitsche et al. [Nitsche and McBride 2018] using a VR controller to manipulate the string puppet. Lin et al. [Lin et al. 2022] designed a mixed-reality puppet show where the audience can use their mobile phones to control virtual puppets on the big screen. AvatarTalk [Luo et al. 2023] even using IoT techniques enables users to control virtual puppets using different sources, like motion capture gloves or camera-based image recognition. Hsieh et al. [Hsieh et al. 2023] utilized an eye-tracker tool to interact with traditional puppetry characters. Way et al. [Way and Wei 2023] used a VR hand tracking system to remotely manipulate the Budaixi puppets simultaneously.

New technologies and puppet manipulation methods are emerging, but most focus only on basic physical movements. They lack recognition of professional traditional puppet manipulation gestures. We aim to create a system that can recognize more advanced hand gestures. The Oculus Quest 2 hand tracking system [Han et al. 2020] allows us to track hands in VR, enabling puppet manipulation through hand movements.

2 DESIGN AND IMPLEMENTATION

The gesture design, developed in collaboration with a Budaixi master, covers the five main character types in traditional Budaixi: shen, dan, jin, chou, and za. Our system maps the hand skeleton data to the puppets' animation (Fig 1.a). The index finger controls the puppet's head, the middle finger controls the puppet's right hand, and the thumb controls the puppet's left hand. Based on the hand skeleton mapping, we developed a gesture recognition system that includes 10 main gestures, where each gesture is accompanied by corresponding visual effects, object interactions, and scene design.

For example, the sleeve-flinging movement for shen (Fig 1.b) is enhanced with line effects to simulate the visualization of a gentle breeze. For the walking movement for dan (Fig 1.c), flower growth and falling effects are added, making the character appear more graceful. The turning movement for jin (Fig 1.d) incorporates

dynamic cape movements and overall puppet rotation, which is a relatively complex movement that becomes easily triggered in VR. The escape movement for chou (Fig 1.e), which in traditional puppetry could only be symbolically represented, can actually be performed in our system, allowing the character to escape from the user's grasp. The backflip movement for za (Fig 1.f), the most challenging movement in reality, requiring extensive practice to master, becomes almost effortless and more interesting in VR, with added effects and interactions making the process even more engaging.

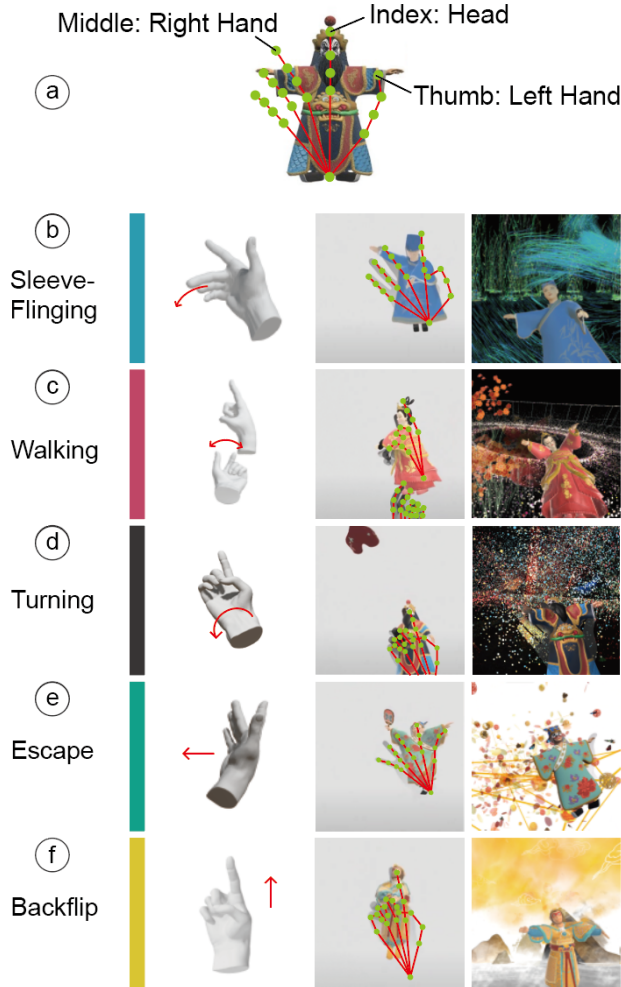


Figure 1: (a)hand skeleton mapping (b)shen gesture (c)dan gesture (d)jin gesture (e)chou gesture (f) za gesture

3 APPLICATION

The experience can be divided into three stages. The first stage (Fig 2.a) features an opening animation with traditional Budaixi objects, a stage, and tools to set the traditional vibe and prepare users for VR. The second stage (Fig 2.b) is a menu where users select a character to manipulate and access gesture tutorials and text information. In

the final stage (Fig 2.c), users use hand gestures to manipulate puppets, triggering visual effects and interactions with virtual objects. The complete experience gives users a more comprehensive understanding, allowing them to fully learn and enjoy our gesture-based puppet manipulation. Among the users, traditional puppet masters offered valuable insights. They highlighted that different schools of Budaixi have varying gesture techniques, showing that virtual puppet manipulation can indeed be a new form of puppetry.

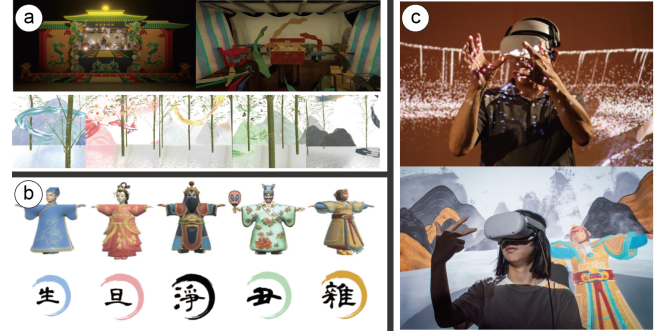


Figure 2: (a) opening animation (b) menu and gesture tutorial (c) main gesture interaction

4 DISCUSSION AND FUTURE WORK

The Theater of Future Puppetry combines an immersive puppeteering experience with hand tracking and gesture recognition to represent traditional Budaixi hand gestures to the public. Our system reveals another possibility for a new form of puppetry. In the future, we will develop a multi-user function to attract more young people to participate in the experience and learn about the culture. We aim to create a new puppet show that offers a unique experience.

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